starlab

_travelers guide
_for deep space exploration

3	4	15
textures	favorites	scales

[infinite] press [infinite] hold › then noon = 12 o'clock cw / ccw = clockwise / counter

time-warped reverberator · reverb, delay, karplus-strong string, shimmer, glimmer, lfo

starlab · 28 hp · manual rev a

violet = hold

01 buttons · LEDs

sl_01

[infinite]	freezes what's going into the reverb · new sound keeps passing through
[favorite]	recalls the selected preset (09)
[regen]	● shimmer inside the reverb, it climbs · off = shimmer on the input only
[high band]	● glimmer lifts the high harmonics · off = the low ones
[echo on]	● the delay joins the wet output · off = it's a pre-delay
[tap / trig]	tap a tempo · karplus: quick press plucks, hold bows
[low pass]	high damp becomes a 24 db resonant low pass on the output

02 reverb

sl_02

[texture]	**sparse** granular · **dense** plate-like · **diffuse** slow, swelling wash
[size / pitch]	cw = bigger room, lower pitch · ±1 octave (½× to 2× size) · retunes the tail
[decay]	how long the tail rings · long settings also feel the filters
[dry]	analog, never digitized · none at min · unity at noon · +3 db at max
[wet]	reverb (and echo) level
[input gain]	into the reverb · LED ● clean › ● soft clip · ● in gate muted

05 harmonics

sl_05

[shimmer]	pitch-shifted reverb · off fully ccw
[regen] + [shimmer]	**interval**: −1 oct to +1 oct · ±10 cent detune either side of noon
[glimmer]	brings out harmonics already in your sound · off fully ccw
[interval] cv	shimmer interval in half steps, ±1 octave

__fig.06 : lfo shapes

[shape]



env follows how loud you play · depth sets its direction, off at noon

__fig.01 : signal path

left to right



echo off = the delay is a pre-delay · infinite holds the input, the reverb keeps running
regen on = shimmer inside the reverb · off = shimmer on the way in

03 delay

sl_03 · delay / karplus

[delay / tune]	0–1.5 s, finer at the short end · with echo off and feedback low it's a pre-delay
[feedback]	more repeats, or resonance at short times
[echo on]	sends the delay to wet beside the reverb
[tap / trig]	tap the time · a clock in the jack turns delay / tune into a ratio (fig.02)
delay only	[decay] at min + [echo on]
longest	7.5 s: a 3 s clock with delay / tune on 5/2

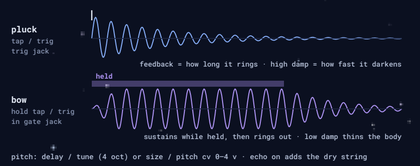
04 karplus-strong

sl_04 · a string voice

[echo on] + [delay] cw	until tap / trig turns ● = string mode · ccw until it blinks ● = delay again
[delay / tune]	tunes the string over 4 octaves · with size cv patched it becomes a 12-semitone fine tune
[tap / trig]	quick = pluck · hold = bow · trig jack plucks, in gate jack bows
[feedback]	how long the string rings
[echo on]	adds the dry string to wet · off = only the reverberated string
pitch cv	[size / pitch] jack, 0–4 v, 1 v/oct

__fig.04 : pluck and bow

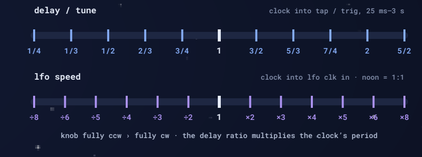
karplus-string mode



pitch: delay / tune (4 oct) or size / pitch cv 0–4 v · echo on adds the dry string

__fig.02 : clocked knobs

what each position means



knob fully ccw › fully cw · the delay ratio multiplies the clock's period

07 lfo

sl_07

[lfo target]	**delay** chorus, flange, vibrato (echo on for more) · **pitch** size / pitch · **filter** high damp (low pass on for sweeps)
[speed]	15 s to 15 hz · env: the decay time
[shape]	triangle · square · ramp · saw · random · env (follows your input)
[depth]	off fully ccw · env: off at noon, ccw negative, cw positive
[lfo clk in]	locks to a clock · speed picks the ratio (fig.02)
[ext lfo in]	your own modulation to the target · depth becomes its attenuverter, off at noon

08 pitch quantize

sl_08 · [infinite]

[infinite]	until it blinks ● = size / pitch steps through a scale · again until ● = off
[infinite] + [size]	pick one of 15 scales · the 4 center buttons show it (fig.05)
[low pass] + [size]	portamento between notes · min = none

__fig.05 : the 15 scales

[infinite] + [size]



binary: tap = 1, echo = 2, band = 4, regen = 8

hold infinite + turn size

* chords build down from a single note

09 favorites

sl_09 · four presets

[favorite]	recall the selected one
[favorite] + [1–4]	choose · the current one shows ●
[favorite] long	until ● › [favorite] saves here, or a center button saves there
[favorite] cv	toggles between the knobs and the preset · dip 1 on = spillover between them

10 cv inputs

sl_10

continuous, ±5 v — the knob is the offset

[size / pitch]	±2 v, 1 v/oct: ±2 octaves · karplus 0–4 v
also	wet · decay · shimmer · interval · glimmer · high damp · feedback · speed · depth · ext lfo

triggers, 0–5 v rising edge

[infinite] [echo on]	toggle · latching by default
[in gate]	mutes the input while high · karplus: bows
[favorite] [clear]	knobs ⟲ preset · empties the reverb
[tap / trig] [lfo clk in]	clocks, 25 ms–3 s · karplus: pluck

06 filter

sl_06 · into the reverb tank

[low damp]	cuts lows in the output and the regenerating core
[high damp]	cuts highs in the regenerating tail
[low pass]	high damp becomes a 24 db / oct low pass on the output
[low pass] + [high damp]	resonance · cw = a sharper peak

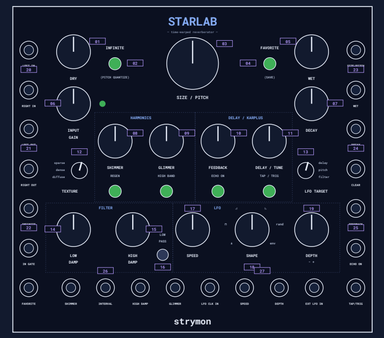
11 the module

sl_11 · 2021

size	28 hp · 41 mm deep · 210 mA on each rail · 96 khz, 32-bit float · analog dry path

__fig.03 : front panel

01–19 = controls · 20–27 = jacks



[01]	__dry : analog dry	[15]	__high damp : highs out
[02]	__infinite : freeze · hold = quantize	[16]	__low pass : 24 db lp · hold = q
[03]	__size / pitch : ±1 oct	[17]	__speed : 15 s–15 hz
[04]	__favorite : presets · hold = save	[18]	__shape : six waves
[05]	__wet : effect level	[19]	__depth : amount · attenuverter
[06]	__input gain : drive · LED	[20]	__ins : left = mono
[07]	__decay : tail length	[21]	__outs : stereo
[08]	__shimmer · regen : pitch shift	[22]	__infinite · gate · fav : triggers
[09]	__glimmer · band : harmonics	[23]	__size · wet · decay : cv
[10]	__feedback · echo : repeats · to wet	[24]	__clear : empty the tail
[11]	__delay · tap : time · tune · pluck	[25]	__fb · echo · tap : cv · clock
[12]	__texture : sparse · dense · diffuse	[26]	__shimmer → glimmer : cv
[13]	__lfo target : delay · pitch · filter	[27]	__lfo jacks : clk · speed · depth · ext
[14]	__low damp : lows out		

_where things hide

no menus · sl_12

held buttons

- infinite pitch quantize · + size = scale
- regen + shimmer shimmer interval
- echo on + delay karplus ‹› delay
- low pass + … high damp = q · size = glide · gain = tracking
- favorite + 1–4 = pick · long = save

at power on

- hold regen infinite cv latch ‹› momentary
- hold high band in gate cv
- hold echo on echo on cv
- dip 1 (back) favorite spillover

LED codes

input gain	● clean ● soft clip ● in gate muted
infinite	● blink = quantize on ● blink = off
tap / trig	● karplus ● blink = delay
favorite	● chosen preset ● flash = save